SCISOFT ACADEMY

OBJECT ORIENTED PROGRAMMING

EPISODE - 5

ENCAPSULATION

Encapsulation:

How do we protect object state?

For example: > BANK ACCOUNT

> balance = 10,000/-

⇩

> balance = -1,00,000/-

Anyone can directly modify balance?

PROBLEM!!!

Instead of this:

User ⟶ balance
Direct

Use this:

User
↓
deposit() / withdraw()
↓
Validation
↓
balance

Encapsulation:

* Control access to an object's internal state through a defined interface.

BANK ACCOUNT

🔒 balance

✓ deposit()
✓ withdraw()
✓ getBalance()

ACCESS MODIFIERS

Note: Access modifier may vary depending on the programming language.

→ Public
→ Private
→ Protected

① Public:

* Accessible through the public interface

② Private:

* Restricted to the class/object implementation

③ Protected:

* Controlled access, commonly relevant to inheritance.

Note: Rules vary by programming language

Validation:

withdraw(₹500)
↓
Is the amount valid? ↓ YES
Is the balance sufficient? ↓ YES → NO ↓ Reject operation
Update balance

Encapsulation helps protect valid state

ENCAPSULATION ≠ GETTERS + SETTERS

* The goal is "CONTROLLED ACCESS" & maintaining valid state.

RECAP

ENCAPSULATION
↓
Hide / Restrict the implementation details
↓
Provide Controlled operations
↓
Protect object invariants / state

What if Car, Bike and Truck share common characteristics?

⇩

INHERITANCE...

NEXT VIDEO